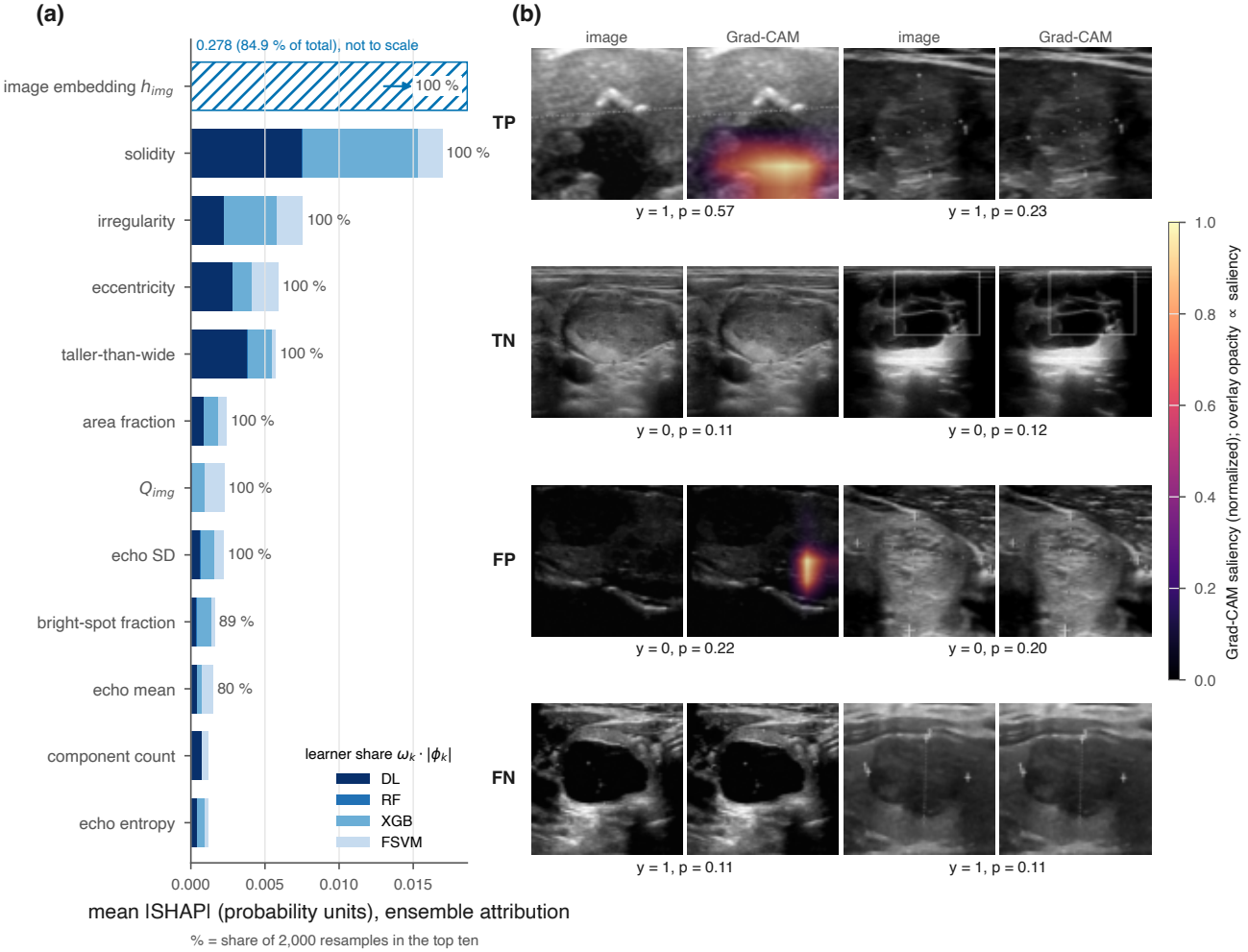

(a)
0.278 (84.9 % of total), not to scale
image embedding $h_{img}$
100 %
solidity
100 %
irregularity
100 %
eccentricity
100 %
taller-than-wide
100 %
area fraction
100 %
$Q_{img}$
100 %
echo SD
100 %
bright-spot fraction
89 %
echo mean
80 %
component count
echo entropy
learner share $\omega_k \cdot |\phi_k|$
DL
RF
XGB
FSVM
0.000
0.005
0.010
0.015
mean ISHAPI (probability units), ensemble attribution
% = share of 2,000 resamples in the top ten
(b)
image
Grad-CAM
image
Grad-CAM
TP
y = 1, p = 0.57
y = 1, p = 0.23
TN
y = 0, p = 0.11
y = 0, p = 0.12
FP
y = 0, p = 0.22
y = 0, p = 0.20
FN
y = 1, p = 0.11
y = 1, p = 0.11
Grad-CAM saliency (normalized); overlay opacity ∝ saliency
1.0
0.8
0.6
0.4
0.2
0.0